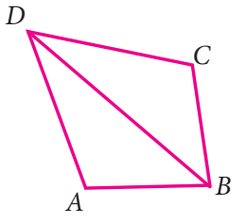
11. Two students drew a line segment each. What is the condition for them to be congruent?
    (i) They should be drawn with a scale.
    (ii) They should be drawn on the same sheet of paper.
    (iii) They should have different lengths.
    (iv) They should have the same length.

12. In the given figure, $AD=CD$ and $AB=CB$. Identify the other three pairs that are equal.



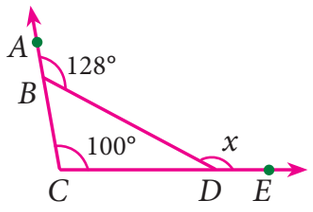
    (i) $\angle ADB = \angle CDB,\ \angle ABD = \angle CBD,\ BD = BD$
    (ii) $AD = AB,\ DC = CB,\ BD = BD$
    (iii) $AB = CD,\ AD = BC,\ BD = BD$
    (iv) $\angle ADB = \angle CDB,\ \angle ABD = \angle CBD,\ \angle DAB = \angle DBC$

13. In $\triangle ABC$ and $\triangle PQR$, $\angle A=50°=\angle P$, $PQ=AB$, and $PR=AC$. By which property $\triangle ABC$ and $\triangle PQR$ are congruent?
    (i) SSS property
    (ii) SAS property
    (iii) ASA property
    (iv) RHS property

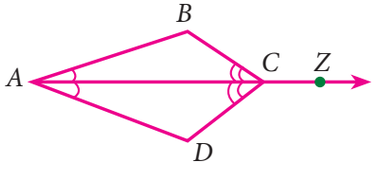
Exercise 4.3

Miscellaneous Practice problems

1. In an isosceles triangle one angle is 76°. If the other two angles are equal find them.
2. If two angles of a triangle are 46° each, how can you classify the triangle?
3. If one angle of a triangle is equal to the sum of the other two angles, find the type of the triangle.
4. If the exterior angle of a triangle is 140° and its interior opposite angles are equal, find all the interior angles of the triangle.
5. In $\triangle JKL$, if $\angle J = 60°$ and $\angle K = 40°$, then find the value of exterior angle formed by extending the side KL.
6. Find the value of 'x' in the given figure.



7. If $\triangle MNO \cong \triangle DEF$, $\angle M = 60°$ and $\angle E = 45°$ then find the value of $\angle O$.
8. In the given figure ray AZ bisects $\angle BAD$ and $\angle DCB$, prove that (i) $\triangle BAC \cong \triangle DAC$ (ii) $AB = AD$.



7th_Maths_T2_EM_Chp4.indd 86 29-05-2019 12:19:43